

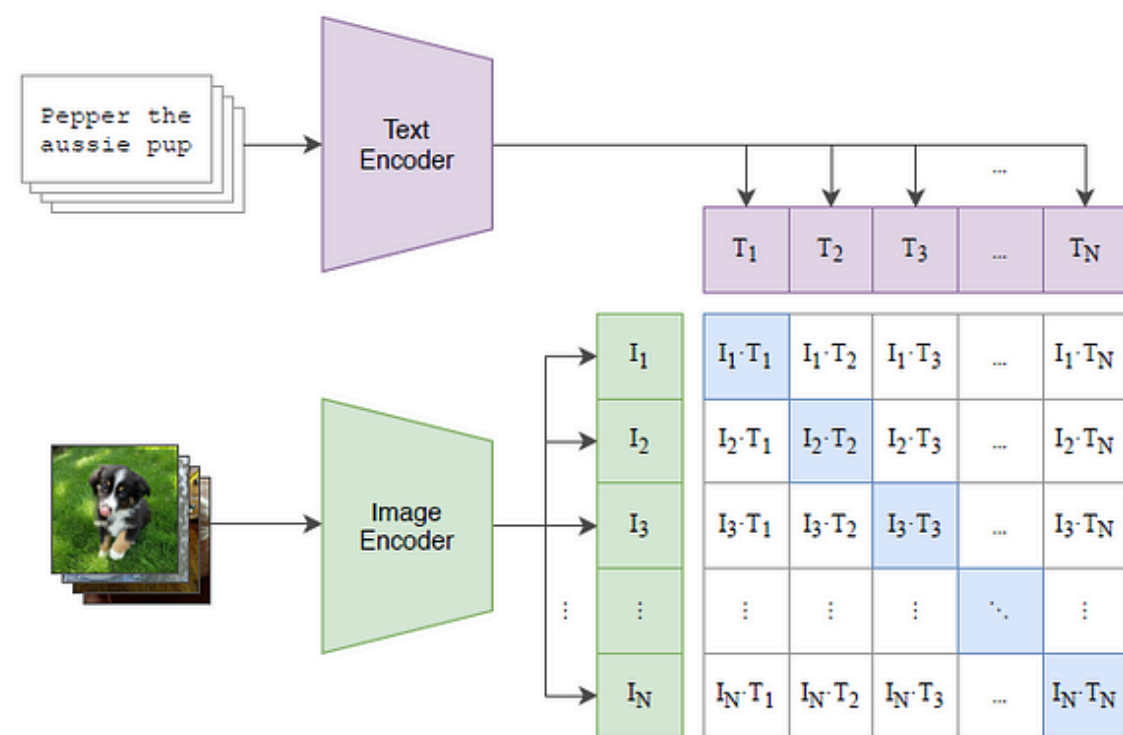
WHEN AND WHY VISION-LANGUAGE MODELS BEHAVE LIKE BAGS-OF-WORDS, AND WHAT TO DO ABOUT IT?

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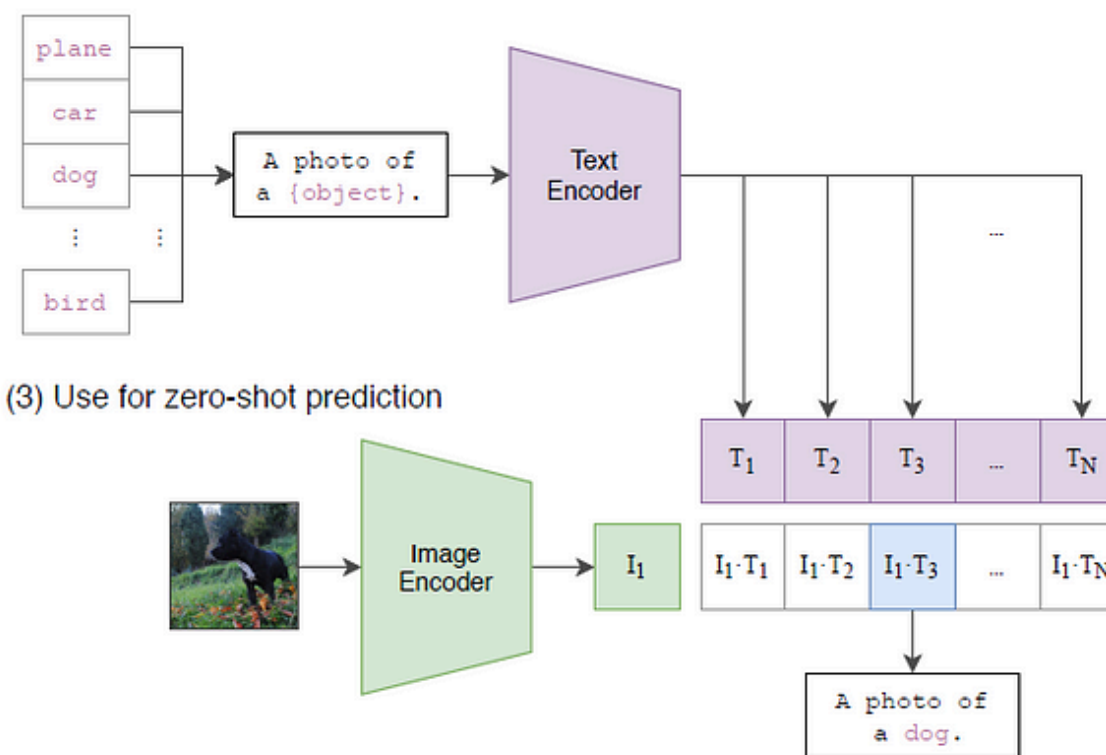
Introduction

Clip

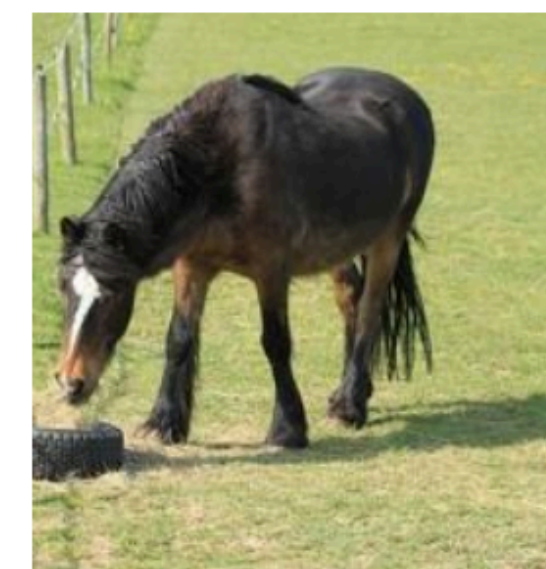
(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



BLIP

the grass is eating the horse 81%

the horse is eating the grass 78%

Plan

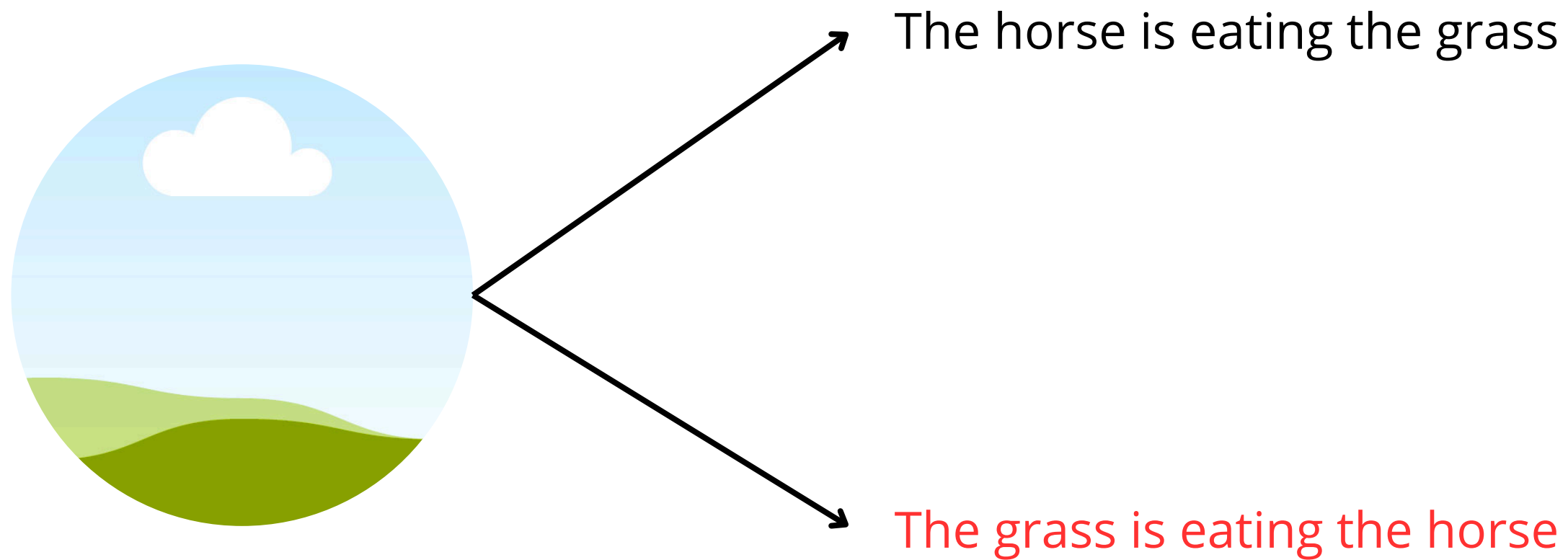
- Reproduction of the VG-R Evaluation

- Reproduction of NegCLIP

- Extension

- Conclusion

Reproduction of VG-R Dataset claim



Assesses the order of a relational expression:
X relation Y vs **Y relation X**

Reproduction of VG-R Dataset claim

Model	Paper	Reproduction
CLIP VIT-B/32	59%	58.37%
BLIP	59%	61.63%

Table 1. Vision-Language models accuracy on VG-R Dataset

Reproduction of VG-R Dataset claim



(b) "the building is behind the man": 0.226
"the man is behind the building": 0.213

Original caption appeared to be ambiguous

Reproduction of NegClip

Examples of hard negatives captions

Original Sentence : Remarkable scene with a blue ball behind a green chair.

Shuffle nouns and adjectives: Green ball with a remarkable chair behind a blue scene.

Shuffle everything but nouns and adjectives: Remarkable scene behind a blue ball with a green chair.

Shuffle trigrams: A green chair remarkable scene with a blue ball behind.

Shuffle words within each trigram: A scene with remarkable a ball blue a green behind chair.

Reproduction of NegClip

Evaluation on Visual Genome Relation and Attribution

Checkpoint	VG-Relation Accuracy	VG-Attribution Accuracy
Epoch 2	81.18%	71.79%
Epoch 5	80.95%	71.38%
Paper Reported	81%	71%

Reproduction of NegClip

Zero-Shot accuracy on CIFAR-10

Model	CIFAR-10
Epoch 2	84.84%
Epoch 5	83.64%
OpenAI CLIP	88.82%
OpenAI CLIP Reported	95%
Paper Reported	94%

Extension



(a) Flux.1-dev

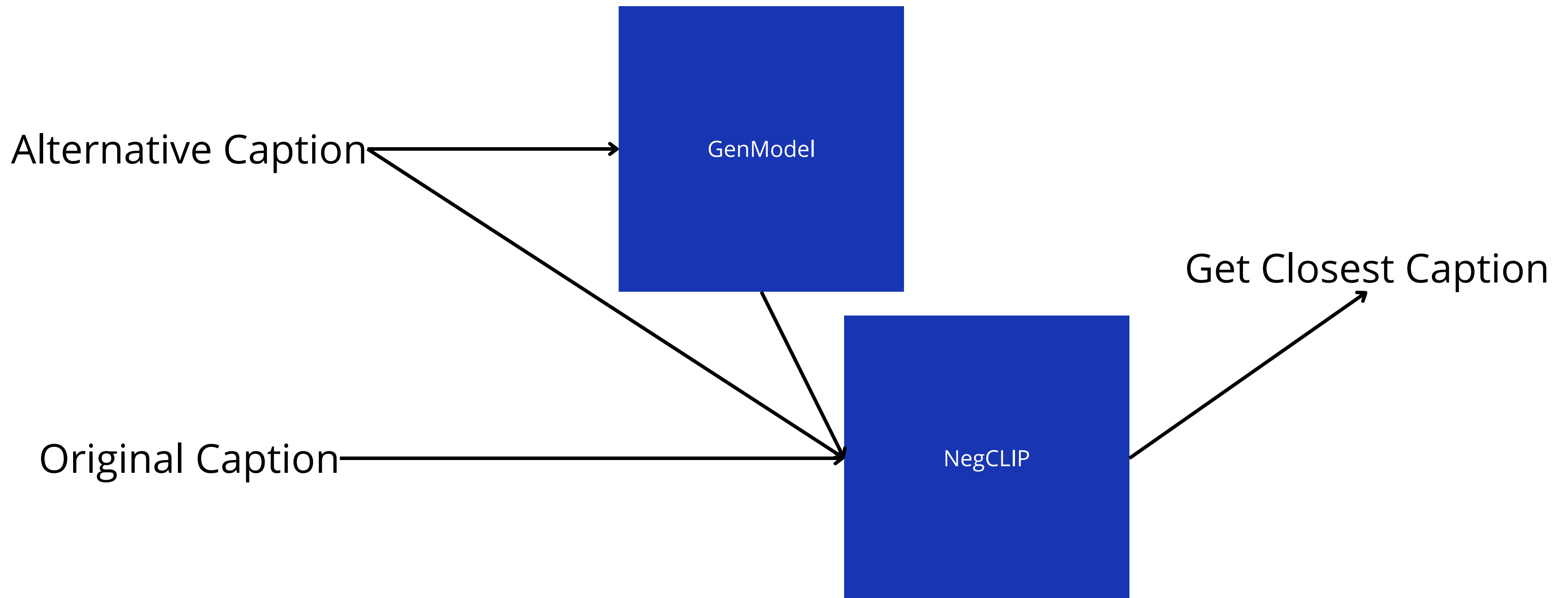


(b) Stable Diffusion v2.1

Figure 3. Image Generated with prompt: "The grass is eating the horse"

Goal : Test whether generative image understand the semantic structure or behave like a simple bag-of-words.

Extension



Extension

Model	Accuracy
Stable Diffusion Turbo XL	7%
stable-diffusion-3.5-large-turbo	4.431%

Table 5. Generative Models accuracy based on NegCLIP similarities

Extension

Issues:

- Poor image quality generated
- Missing the evaluation of the image quality and similarity compared to the original image
- NegCLIP is not perfect
- Our prompting approach is not perfect leading to not perfect image too



(b) Image for *"the shirt is wearing the man"*

Conclusion

Contributions:

- Reproduction of a benchmark (VG-R) on two Visual Language Models
- Training and reproduction of NegCLIP
- Proposed a benchmark on evaluating prompt alignment of Image Generative Models using the paper contributions

Future Works:

- Take a look at newer visual-text alignment dataset (CoBSAT)
- Experiment on newer Visual Language Models
- Improve our benchmark on both tested models and stated limitations

Conclusion

Task	Member
Reproduce Benchmark	Victor Guichard
Reproduce NegCLIP	Eliott Vigier
Implement Extension	Paul Chu, Victor Guichard

Merci !